# Causal Phase Obstructions in Exact Information Simulation

Zero-Repair Trees, Hidden Ordering Information, and Multiscale Renormalization

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**Abstract**

We study an exact causal information-accounting problem in which two finite information-increment laws must be reconciled by a nonnegative, history-dependent inventory, with no fresh repair term. The local constraint is a linear convolution identity, but the global feasibility problem is strongly order-sensitive.

Our first result is a finite-horizon converse. For every fixed Laplace tilt we derive an explicit logarithmic lower bound for the minimum root inventory $\mathcal{B}_n$. If the increment laws have equal mean and variance and their first mismatch is an oriented third cumulant, then

$$\liminf_{n\to\infty} \frac{\mathcal{B}_n}{\log n} \geq \frac{\Delta\kappa_3}{6\sigma^2}.$$

For the dyadic pair

$$\mu_P = \tfrac{1}{4}\delta_2 + \tfrac{3}{4}\delta_4, \qquad \mu_Q = \tfrac{3}{4}\delta_3 + \tfrac{1}{4}\delta_5,$$

this gives the exact lower-bound coefficient 1/3 and rules out every infinite finite-mean zero-repair tree.

We then isolate a second obstruction which is invisible to scalar information spectra. At block length $m$, ordered binary histories form a space $V_m \cong \mathbb{R}^{2^m}$, while scalar total information sees only the $m+1$ Hamming layers. The lost ordering information is the phase space $W_m = \ker \Pi_m$, of dimension

$$\dim W_m = 2^m - (m+1).$$

Under block doubling it has the exact decomposition

$$W_{2m} = (W_m \otimes W_m) \oplus (W_m \otimes U_m) \oplus (U_m \otimes W_m) \oplus K_m, \qquad \dim K_m = m^2,$$

where $K_m$ is a fresh anti-diagonal phase sector created by coarse-graining. For two symbols we solve the associated causal matching LP exactly: the minimum information-profile mismatch is 5/16, whereas static matching pays only 1/4. An explicit primal coupling and an explicit dual certificate prove optimality. The extra 1/16 is added to both scalar residual profiles and therefore disappears from their signed difference. Causality can thus create a real obstruction living entirely in the kernel of scalar information projection.

Finally, we pass from viability sets to lower support functionals, derive an exact finite-dimensional Bellman–Fenchel recursion, identify the cubic shift normal form

$$P^* - Q^* = \tfrac{1}{4}E^2(I-E)^3,$$

and introduce a Green coordinate in which the cubic defect becomes a positive unit source. The two-block boundary mask is $(1+z)^3/4$, the binary quadratic B-spline mask, and one explicit mean-neutral phase channel admits a uniform finite-mean envelope across all dyadic scales. These statements do not prove a matching $O(\log n)$ upper bound. They reduce the sharp problem to the renormalization of a positive causal phase fiber. In particular, a uniform bound on $\mathcal{B}_{2n} - \mathcal{B}_n$ would close the order of growth.

The zero-repair bill is deliberately distinguished from unrestricted causal Shannon entropy. It measures exact causal working inventory, not entropy rate; this distinction is the link back to the broader question of which resources an exact stochastic realization must carry.

**Keywords:** causal information, exact simulation, zero repair, causal transport, information spectrum, renormalization, Bellman–Fenchel duality, subdivision schemes, third cumulant.

# 1 Introduction

A probability law tells us what may be observed. It does not, by itself, tell us how much hidden state an exact realization must carry when outputs are chosen sequentially and future repair is forbidden. This paper isolates one such resource question.

Suppose two actions have prescribed information-increment laws $\mu_P$ and $\mu_Q$. At every history we ask for an exact accounting identity: the current nonnegative inventory plus the next $Q$-increment must have the same law as a $P$-branch increment plus the next inventory assigned to that branch. No signed debt is allowed and no fresh repair term may be injected. The resulting object is an exact zero-repair information tree.

There are two reasons to study this stricter contract separately from ordinary entropy-rate simulation. First, exact online random generation can recycle unused randomness and operate arbitrarily close to Shannon entropy in models that allow a persistent state and fresh source bits [13, 14]. Second, finitary coding theory shows that equal entropy alone does not control stronger moments of coding radius or information cocycles [2, 3, 4]. Thus rate and working state are different resources.

The present problem also sits near, but not inside for free, modern adapted optimal transport. Causal and bicausal transport admit general duality and dual attainment under broad assumptions [9]; graph-causal transport now provides gluing and dynamic-programming results for couplings constrained by directed acyclic graphs [16]. Our model has an additional exact conservation law and a hard state constraint $J \geq 0$. These are not cosmetic: they are the source of the obstruction.

A useful recent methodological parallel comes from synchronization channels. Kovačević identifies a coefficientwise domination condition for a family of sticky channels, converts it into an exact dual capacity bound, and then characterizes the laws satisfying that condition through a functional composition equation [15]. The present setting is different, but the same standard is appropriate: the objective is not another successful ansatz, but an exact structural criterion exposing the admissible positive fiber on which the dynamics acts.

## 1.1 Main contributions

The paper has six main components.

1. We give a self-contained finite-horizon logarithmic converse for exact zero repair. The third-cumulant case yields an explicit lower-bound coefficient, equal to $1/3$ for the dyadic parity pair.
2. We represent finite viability sets by their lower support functionals and derive an exact Bellman–Fenchel recursion in finite dimensions, together with a rigorous weak-duality version on the countable state space.
3. We introduce the ordered-history phase quotient. Scalar total information is a projection $\Pi_m$ from $2^m$ ordered histories to $m+1$ Hamming layers; its kernel has dimension $2^m - (m+1)$.
4. We solve the first nontrivial causal phase LP exactly. The two-symbol causal mismatch is $5/16$, compared with the static value $1/4$, and we give exact primal and dual certificates.
5. We prove a block-doubling decomposition of the phase kernel. Besides inherited phase directions, each doubling creates a fresh $m^2$-dimensional anti-diagonal sector. This rules out any globally one-dimensional phase picture.
6. We identify the cubic finite-difference normal form, a Green-coordinate affine recursion, the associated quadratic B-spline boundary mask, and one explicit marginal phase channel

that can be transported through all dyadic scales at bounded mean cost. The sharp upper bound remains open; the paper ends with a precise renormalization target rather than a claimed QED.

### 1.2 Resource interpretation and scope

The quantity studied here is not automatically the excess Shannon entropy of an unrestricted causal simulator. It is an auxiliary working-inventory cost under a stricter contract. We will return to this distinction in Section 17. In particular, no ontological conclusion is drawn from the mathematics: the paper prices a stated realization constraint.

## 2 Exact zero-repair information trees

Let

$$\mu_P = \sum_{i=1}^{r} p_i \delta_{x_i}, \qquad p_i > 0, \qquad \sum_i p_i = 1,$$

and let $\mu_Q$ be a finite-support probability law on $[0, \infty)$. Branch labels are retained even if two values $x_i$ coincide; equal current information does not imply equal future subtrees.

**Definition 2.1** (Depth-$n$ exact zero-repair tree)**.** A depth-$n$ zero-repair tree is a family of probability laws

$$\{B_h\}_{|h|\leq n}$$

on $[0, \infty)$, indexed by histories $h \in \{1, \dots, r\}^{\leq n}$, such that each internal node satisfies

$$\boxed{\mu_Q * B_h = \sum_{i=1}^{r} p_i \, \delta_{x_i} * B_{hi}.} \tag{1}$$

For a buffer law $B_h$, put $m_h := \mathbb{E}_{B_h} J$.

**Definition 2.2** (Finite-horizon zero-repair bill)**.**

$$\boxed{\mathcal{B}_n := \inf \left\{ m_\varnothing : \{B_h\}_{|h|\leq n} \text{ satisfies (1)} \right\}.} \tag{2}$$

The value is $+\infty$ if no feasible tree exists.

Throughout the converse we assume equal increment means,

$$\mathbb{E}_{\mu_P} X = \mathbb{E}_{\mu_Q} Y. \tag{3}$$

**Lemma 2.3** (Mean martingale)**.** *If $m_h < \infty$, then all child means are finite and*

$$\boxed{m_h = \sum_i p_i m_{hi}.} \tag{4}$$

*Consequently, along an independently $P$-distributed branch history, $m_{H_t}$ is a nonnegative martingale.*

*Proof.* Take first moments in (1). Finite support and nonnegative child means imply finiteness of each child mean. The common increment mean cancels, leaving (4). ☐

*Remark* 2.4 (Constrained coboundary form)*.* After coupling the equal laws in (1), one may write schematically

$$J_h + Y = x_I + J_{hI}.$$

Thus $Y - x_I = J_{hI} - J_h$ resembles a coboundary. The difficulty is the simultaneous imposition of nonnegativity, exact branch weights, convolutional input, and future viability at every history.

## 3 A finite-horizon logarithmic converse

Fix $a > 0$ and define

$$G_P(a) := \mathbb{E}_{\mu_P} e^{-aX}, \qquad G_Q(a) := \mathbb{E}_{\mu_Q} e^{-aY}, \qquad \rho(a) := \frac{G_Q(a)}{G_P(a)}.$$

Introduce the tilted branch law

$$\widetilde{p}_i^{(a)} := \frac{p_i e^{-ax_i}}{G_P(a)}$$

and its likelihood-ratio second moment

$$\boxed{\Lambda(a) := \sum_i \frac{(\widetilde{p}_i^{(a)})^2}{p_i} = \frac{G_P(2a)}{G_P(a)^2} \geq 1.} \tag{5}$$

For each node set

$$\widehat{B}_h(a) := \mathbb{E}_{B_h} e^{-aJ}, \qquad y_h := e^{-am_h}.$$

By Jensen and $J \geq 0$,

$$0 < y_h \leq \widehat{B}_h(a) \leq 1. \tag{6}$$

For a $k$-step descendant $hw$, define

$$D_h^{(k)} := \mathbb{E}_{P^k} y_{hw} - y_h.$$

**Lemma 3.1** (Jensen–Hellinger control)**.** *For every node and every admissible $k$,*

$$D_h^{(k)} \geq 0,$$

*and*

$$\mathbb{E}_{P^k} \left(\sqrt{y_{hw}} - \sqrt{y_h}\right)^2 \leq D_h^{(k)}, \qquad \mathbb{E}_{P^k} (y_{hw} - y_h)^2 \leq 4D_h^{(k)}. \tag{7}$$

*Proof.* By (4), $\mathbb{E}_{P^k} m_{hw} = m_h$. Convexity of $u \mapsto e^{-au}$ gives $D_h^{(k)} \geq 0$, while

$$\mathbb{E} e^{-am_{hw}/2} \geq e^{-a\mathbb{E} m_{hw}/2} = \sqrt{y_h}.$$

Expanding the square proves the first estimate. Since $|u - v| \leq 2|\sqrt{u} - \sqrt{v}|$ on $[0, 1]$, the second follows. $\square$

Taking Laplace transforms in (1) yields the exact tilted identity

$$\boxed{\mathbb{E}_{\widetilde{P}_a} \widehat{B}_{hi}(a) = \rho(a) \widehat{B}_h(a),} \qquad \boxed{\mathbb{E}_{\widetilde{P}_a^k} \widehat{B}_{hw}(a) = \rho(a)^k \widehat{B}_h(a).} \tag{8}$$

Let

$$L_w := \frac{\widetilde{P}_a^k(w)}{P^k(w)}.$$

Then $\mathbb{E} L_w = 1$ and $\mathbb{E} L_w^2 = \Lambda(a)^k$.

**Lemma 3.2** (One-block measure change)**.** *For every node $h$,*

$$\boxed{\rho(a)^k \widehat{B}_h(a) \geq y_h - 2\Lambda(a)^{k/2} \sqrt{D_h^{(k)}}.} \tag{9}$$

*Proof.* Since $\mathbb{E}(L_w - 1) = 0$,

$$\mathbb{E}_{\widetilde{P}_a^k} y_{hw} = y_h + \mathbb{E}_{P^k} [(L_w - 1)(y_{hw} - y_h)].$$

Cauchy–Schwarz and (7) give the stated lower bound. Finally use $\widehat{B}_{hw}(a) \geq y_{hw}$ and (8). $\square$

**Proposition 3.3** (Master inequality)**.** *Let a feasible depth-$n$ tree have root mean $m$. For every integer $k \le n/2$,*

$$\boxed{\rho(a)^k \ge e^{-am} - 2\Lambda(a)^{k/2}\sqrt{\frac{2amk}{n}}.} \tag{10}$$

*Proof.* Put $q = \lfloor n/k \rfloor \ge n/(2k)$. The disjoint $k$-blocks telescope:

$$\sum_{j=0}^{q-1} \mathbb{E}D^{(k)}_{H_{jk}} = \mathbb{E}y_{H_{qk}} - y_{\varnothing} \le 1 - e^{-am} \le am.$$

Hence one block start satisfies

$$\mathbb{E}D^{(k)}_{H_{jk}} \le \frac{2amk}{n}.$$

Average (9) over that starting history, use $\mathbb{E}m_{H_{jk}} = m$, Jensen, $\widehat{B}_h \le 1$, and $\mathbb{E}\sqrt{D} \le \sqrt{\mathbb{E}D}$. □

**Theorem 3.4** (Fixed-tilt logarithmic obstruction)**.** *Assume* (3)*. If $\rho(a) < 1$ for some fixed $a > 0$, set*

$$d(a) := -\log\rho(a) > 0, \qquad \Gamma(a) := 2a + \frac{a\log\Lambda(a)}{d(a)}.$$

*Then*

$$\boxed{\liminf_{n\to\infty} \frac{\mathcal{B}_n}{\log n} \ge \frac{1}{\Gamma(a)}.} \tag{11}$$

*Here and below* log *denotes the natural logarithm.*

*Proof.* For a feasible depth-$n$ tree of root mean $m$, choose

$$k = \left\lceil \frac{am + \log 2}{d(a)} \right\rceil.$$

Then $\rho(a)^k \le \frac{1}{2}e^{-am}$. If $k > n/2$, the definition of $k$ already gives a lower bound on $m$ linear in $n$, stronger than (11). Otherwise combine this estimate with (10) and square to obtain

$$n \le 32amk\,\Lambda(a)^k e^{2am}.$$

The ceiling bound on $k$ therefore yields a finite constant $C_a$ such that

$$n \le C_a(m+1)^2 e^{\Gamma(a)m}.$$

Taking logarithms,

$$\log n \le \Gamma(a)m + 2\log(m+1) + \log C_a.$$

Apply this inequality to a sequence of feasible trees whose root means are within $1/n$ of $\mathcal{B}_n$, and pass to a subsequence realizing the liminf. If the ratio is finite, then $m = O(\log n)$ on that subsequence and $\log(m+1) = o(\log n)$. Division by $\log n$ proves (11). □

**Theorem 3.5** (Third-cumulant logarithmic obstruction)**.** *Assume that $\mu_P$ and $\mu_Q$ have common mean $\mu$, common variance $\sigma^2 > 0$, and*

$$\Delta\kappa_3 := \kappa_3(\mu_Q) - \kappa_3(\mu_P) > 0.$$

*Then*

$$\boxed{\liminf_{n\to\infty} \frac{\mathcal{B}_n}{\log n} \ge \frac{\Delta\kappa_3}{6\sigma^2}.} \tag{12}$$

*In particular $\mathcal{B}_n \to \infty$.*

*Proof.* Let
$$K_P(a) := \log \mathbb{E}e^{-aX}, \qquad K_Q(a) := \log \mathbb{E}e^{-aY}.$$
Finite support gives, as $a \downarrow 0$,
$$K_P(a) = -\mu a + \frac{\sigma^2}{2}a^2 - \frac{\kappa_3(P)}{6}a^3 + O(a^4),$$
$$K_Q(a) = -\mu a + \frac{\sigma^2}{2}a^2 - \frac{\kappa_3(Q)}{6}a^3 + O(a^4).$$
Hence
$$d(a) = K_P(a) - K_Q(a) = \frac{\Delta\kappa_3}{6}a^3 + O(a^4),$$
and
$$\log \Lambda(a) = K_P(2a) - 2K_P(a) = \sigma^2 a^2 + O(a^3).$$
Therefore
$$\Gamma(a) \longrightarrow \frac{6\sigma^2}{\Delta\kappa_3}.$$
Apply Theorem 3.4 for each sufficiently small fixed $a > 0$, first take $n \to \infty$, and only then send $a \downarrow 0$. □

# 4 The dyadic parity pair

We now specialize to
$$\boxed{\mu_P = \frac{1}{4}\delta_2 + \frac{3}{4}\delta_4, \qquad \mu_Q = \frac{3}{4}\delta_3 + \frac{1}{4}\delta_5.} \tag{13}$$
Both laws have mean $7/2$, variance $3/4$, and centered third cumulants
$$\kappa_3(P) = -\frac{3}{4}, \qquad \kappa_3(Q) = +\frac{3}{4}.$$
Thus $\Delta\kappa_3 = 3/2$.

**Corollary 4.1** (The 1/3 lower law)**.** *For* (13),
$$\boxed{\liminf_{n\to\infty} \frac{\mathcal{B}_n}{\log n} \geq \frac{1}{3}.} \tag{14}$$
*Equivalently,*
$$\mathcal{B}_n \geq \left(\frac{1}{3} - o(1)\right) \log n.$$
*For base-two logarithms the coefficient is* $(\log 2)/3$*.*

With $z = e^{-a}$, the increment PGFs are
$$G_P(z) = \frac{1}{4}z^2 + \frac{3}{4}z^4, \qquad G_Q(z) = \frac{3}{4}z^3 + \frac{1}{4}z^5,$$
and
$$\boxed{G_P(z) - G_Q(z) = \frac{1}{4}z^2(1-z)^3.} \tag{15}$$

**Corollary 4.2** (No finite-mean infinite zero-repair tree)**.** *There is no infinite history-indexed zero-repair tree for* (13) *with finite root mean.*

*Proof.* Such a tree would restrict to a feasible depth-$n$ tree for every $n$, forcing $\mathcal{B}_n$ to be uniformly bounded by its root mean, contrary to (14). □

## 5 Viability kernels and Bellman–Fenchel duality

For the dyadic pair, write the probability generating function of a buffer law as the same symbol. Multiplying (1) by four gives

$$\boxed{B_{h0}(z) + 3z^2 B_{h1}(z) = A(z)B_h(z), \qquad A(z) := z(3 + z^2).} \tag{16}$$

Define

$$\mathcal{K}_0 := \mathcal{P}(\mathbb{Z}_{\geq 0})$$

and recursively

$$\mathcal{K}_{n+1} := \{B \in \mathcal{P} : \exists B_0, B_1 \in \mathcal{K}_n \text{ satisfying (16)}\}\,. \tag{17}$$

Then

$$\mathcal{B}_n = \inf_{B \in \mathcal{K}_n} \mathbb{E}_B J.$$

**Proposition 5.1** (Convexity)**.** *Every $\mathcal{K}_n$ is convex.*

*Proof.* The base $\mathcal{K}_0$ is convex. If $B, B' \in \mathcal{K}_{n+1}$ have feasible child pairs $(B_0, B_1)$ and $(B_0', B_1')$, then linearity of (16) shows that every convex combination of $B, B'$ has the corresponding convex combinations of the children, which lie in $\mathcal{K}_n$. □

For a convex set $K$ of probability laws and a test function $j$, define its lower support functional by

$$\ell_K(j) := \inf_{B \in K} \langle j, B \rangle. \tag{18}$$

Write $\ell_n := \ell_{\mathcal{K}_n}$. Then

$$\mathcal{B}_n = \ell_n(\mathrm{id}), \qquad \mathrm{id}(t) = t.$$

On a finite state space, lower support is a complete coordinate for a closed convex viability set.

**Proposition 5.2** (Lossless finite-dimensional support representation)**.** *Let $X$ be finite, $\Delta_X$ the probability simplex, and $K \subseteq \Delta_X$ nonempty, closed and convex. Then*

$$K = \bigcap_{j \in \mathbb{R}^X} \{b \in \Delta_X : \langle j, b \rangle \geq \ell_K(j)\}. \tag{19}$$

*Proof.* The forward inclusion is immediate. If $b \notin K$, finite-dimensional separation gives $j$ such that $\langle j, b \rangle < \inf_{x \in K} \langle j, x \rangle$. □

We now state the abstract dual recursion. Let $X, Y$ be finite sets, $K \subseteq \Delta_X$ nonempty compact convex, and $C, S_1, \ldots, S_r : \mathbb{R}^X \to \mathbb{R}^Y$ linear. For positive weights $p_i$ with $\sum_i p_i = 1$, define

$$\Phi(K) := \left\{ b \in \Delta_X : \exists b_i \in K, \quad Cb = \sum_i p_i S_i b_i \right\}.$$

**Theorem 5.3** (Bellman–Fenchel duality)**.** *If $\Phi(K) \neq \varnothing$, then for every $j \in \mathbb{R}^X$,*

$$\boxed{\ell_{\Phi(K)}(j) = \sup_{\lambda \in \mathbb{R}^Y} \left\{ \min_{x \in X} (j_x - (C^*\lambda)_x) + \sum_i p_i \ell_K(S_i^* \lambda) \right\}.} \tag{20}$$

*Proof.* Consider the convex program minimizing $\langle j, b \rangle$ over $(b, b_1, \ldots, b_r) \in \Delta_X \times K^r$ subject to the displayed linear equality. With multiplier $\lambda$, the Lagrangian is

$$\langle j - C^*\lambda, b \rangle + \sum_i p_i \langle S_i^* \lambda, b_i \rangle.$$

Minimizing over the product set gives the right-hand side of (20). The primal feasible set is compact and convex and the constraints are linear, so finite-dimensional strong duality follows from separation [1]. □

For the dyadic model,

$$(C_Q^*\lambda)(t) = \frac{3}{4}\lambda(t+3) + \frac{1}{4}\lambda(t+5),$$
$$(S_2^*\lambda)(t) = \lambda(t+2), \qquad (S_4^*\lambda)(t) = \lambda(t+4).$$

Thus the formal countable-state Bellman transform is

$$\boxed{\begin{aligned}(\mathfrak{D}\ell)(j) = \sup_{\lambda}\Bigg\{ &\inf_{t\ge 0}\left[j(t) - \frac{3}{4}\lambda(t+3) - \frac{1}{4}\lambda(t+5)\right] \\ &+ \frac{1}{4}\ell(\lambda(\cdot+2)) + \frac{3}{4}\ell(\lambda(\cdot+4))\Bigg\}.\end{aligned}} \tag{21}$$

On a finite exact truncation, $\ell_{n+1} = \mathfrak{D}\ell_n$. On the full countable state space, the same expression is always a valid lower certificate for each admissible $\lambda$, but equality requires a separate no-gap theorem in a suitable weighted function space.

**Proposition 5.4** (Weak countable-state recursion)**.** *For every admissible $\lambda$ for which the pairings are finite,*

$$\begin{aligned}\ell_{n+1}(j) \ge &\inf_{t\ge 0}\left[j(t) - \frac{3}{4}\lambda(t+3) - \frac{1}{4}\lambda(t+5)\right] \\ &+ \frac{1}{4}\ell_n(\lambda(\cdot+2)) + \frac{3}{4}\ell_n(\lambda(\cdot+4)).\end{aligned} \tag{22}$$

*Proof.* For any feasible parent and children, add the zero pairing of $\lambda$ with the constraint. Bound the parent contribution by its pointwise infimum and each child contribution by the corresponding lower support, then take the infimum over feasible triples. ☐

The operator is order preserving and additively homogeneous on every domain where the expressions are finite:

$$\ell_1 \le \ell_2 \Rightarrow \mathfrak{D}\ell_1 \le \mathfrak{D}\ell_2, \qquad \mathfrak{D}(\ell + c) = \mathfrak{D}\ell + c.$$

Consequently, on any invariant class on which the uniform norm is finite,

$$\|\mathfrak{D}\ell_1 - \mathfrak{D}\ell_2\|_\infty \le \|\ell_1 - \ell_2\|_\infty.$$

There is also the exact multiplier gauge $\lambda \sim \lambda + c$, a direct consequence of probability normalization. These are the structural features of a topical or Shapley-type operator; they motivate, but do not by themselves justify, nonlinear Perron–Frobenius methods [5, 8, 7].

# 6 Ordered histories and the phase quotient

The scalar information spectrum forgets order. The causal tree does not. We now make the difference algebraic.

Let

$$V_m := \mathbb{R}^{\{0,1\}^m}$$

with canonical basis $e_w$, $w \in \{0,1\}^m$. For a binary word $w$, let $\mathrm{wt}(w)$ denote its Hamming weight. Define the scalarization map

$$\boxed{\Pi_m : V_m \to \mathbb{R}^{m+1}, \qquad \Pi_m e_w = e_{\mathrm{wt}(w)}.} \tag{23}$$

For a signed measure on ordered histories, $\Pi_m$ simply sums its masses within equal-Hamming-weight layers. Since the total information of either binary source is an affine function of Hamming weight, this is exactly the projection from ordered histories to scalar total-information profiles.

**Definition 6.1** (Ambient phase space)**.**

$$\boxed{W_m := \ker \Pi_m.} \tag{24}$$

Elements of $W_m$ are signed rearrangements invisible to scalar total information.

**Theorem 6.2** (Dimension of hidden ordering information)**.** *For every $m \geq 1$,*

$$\boxed{\dim W_m = 2^m - (m+1).} \tag{25}$$

*Proof.* The map $\Pi_m$ is surjective: for every $0 \leq k \leq m$, any word of weight $k$ maps to the $k$-th basis vector. Hence $\operatorname{rank} \Pi_m = m+1$, and rank-nullity gives (25). □

Define normalized layer vectors

$$u_k^{(m)} := \binom{m}{k}^{-1} \sum_{\mathrm{wt}(w)=k} e_w, \qquad 0 \leq k \leq m,$$

and let

$$U_m := \operatorname{span}\{u_0^{(m)}, \ldots, u_m^{(m)}\}.$$

Then $\Pi_m|_{U_m}$ is an isomorphism and

$$V_m = U_m \oplus W_m. \tag{26}$$

Thus $U_m$ is a canonical visible sector and $W_m$ the invisible phase sector.

For $m = 2$,

$$u_0 = e_{00}, \qquad u_1 = \frac{1}{2}(e_{01} + e_{10}), \qquad u_2 = e_{11},$$

and

$$W_2 = \operatorname{span}\{\theta\}, \qquad \theta := \frac{1}{2}(e_{01} - e_{10}). \tag{27}$$

This is the first nontrivial ordering phase: the words 01 and 10 have the same scalar information but different causal pasts.

## 6.1 Two-block phase coordinates for buffer PGFs

Iterating (16) twice gives

$$\boxed{B_{00} + qB_{01} + qB_{10} + q^2 B_{11} = A^2 B, \qquad q := 3z^2.} \tag{28}$$

Put

$$S := \frac{B_{01} + B_{10}}{2}, \qquad \Theta := \frac{B_{01} - B_{10}}{2}. \tag{29}$$

Then the coarse equation loses $\Theta$:

$$\boxed{B_{00} + 2qS + q^2 B_{11} = A^2 B.} \tag{30}$$

**Proposition 6.3** (Exact two-block lifting criterion)**.** *Let $B, B_{00}, S, B_{11}$ be probability PGFs satisfying* (30)*. They admit a genuine two-step zero-repair lift if and only if there exists a signed series $\Theta$ with $\Theta(1) = 0$ such that*

$$B_{01} = S + \Theta, \qquad B_{10} = S - \Theta$$

*are probability PGFs and*

$$\boxed{B_0 = \frac{B_{00} + q(S + \Theta)}{A}, \qquad B_1 = \frac{S - \Theta + qB_{11}}{A}} \tag{31}$$

*are probability PGFs.*

*Proof.* Necessity follows by taking $S, \Theta$ from an existing two-step tree and solving the first-level recursions for $B_0, B_1$. Conversely, if the four PGFs in the statement are nonnegative and normalized, the formulas imply

$$B_{00} + qB_{01} = AB_0, \qquad B_{10} + qB_{11} = AB_1,$$

and adding $B_0 + qB_1 = AB$ is equivalent to (30). Hence the full two-step tree closes exactly. $\square$

*Remark* 6.4. Proposition 6.3 is a change of coordinates, not a solution of the positivity problem. Its value is that it isolates the missing datum: the coarse scalar equation determines the visible part, while causal feasibility asks whether the corresponding phase fiber contains a point compatible with all PGF inequalities.

# 7 An exact two-symbol causal phase surcharge

The phase quotient is not merely formal. The first nontrivial block already contains an obstruction that is absent from static scalar matching.

Let $X = (X_1, X_2) \sim P^{\otimes 2}$ and $W = (W_1, W_2) \sim Q^{\otimes 2}$, where

$$P(0) = \frac{1}{4}, \quad P(1) = \frac{3}{4}, \qquad Q(0) = \frac{3}{4}, \quad Q(1) = \frac{1}{4}.$$

The two-symbol total-information levels are

$$I_P(00, 01, 10, 11) = (4, 6, 6, 8),$$

$$I_Q(00, 01, 10, 11) = (6, 8, 8, 10).$$

A coupling $\pi$ of $P^{\otimes 2}$ and $Q^{\otimes 2}$ is causal from $W$ to $X$ if the conditional law of $X_1$ given $(W_1, W_2)$ depends only on $W_1$. In finite form, for all $a, b, c \in \{0, 1\}$,

$$\sum_{x_2} \pi((a, x_2), (b, c)) = Q(c) \sum_{x_2, w_2} \pi((a, x_2), (b, w_2)). \tag{32}$$

Define the mismatch cost

$$c(x, w) := \mathbf{1}\{I_P(x) \neq I_Q(w)\}.$$

**Theorem 7.1** (Exact two-symbol causal mismatch)**.** *Among all causal couplings $\pi$ satisfying the prescribed product marginals,*

$$\boxed{\min_{\pi} \mathbb{E}_\pi c(X, W) = \frac{5}{16}.} \tag{33}$$

*Equivalently, the maximum causally matchable equal-information mass is $11/16$. Without the causal constraint, the minimum mismatch is $1/4$, so causality creates an additional surcharge $1/16$.*

*Proof.* Order rows and columns as $00, 01, 10, 11$. The matrix

$$\pi = \frac{1}{16} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 3 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 \\ 3 & 2 & 3 & 1 \end{pmatrix} \tag{34}$$

has row sums $(1, 3, 3, 9)/16 = P^{\otimes 2}$, column sums $(9, 3, 3, 1)/16 = Q^{\otimes 2}$, and satisfies (32). Its mismatch mass is $5/16$, proving the upper bound.

For the reverse inequality, write the equality constraints in the order: four $X$-marginals, four $W$-marginals, then the eight causal equations (32) in lexicographic order of $(a, b, c)$. The dual vector

$$y = \left(1, 0, -\frac{1}{4}, \frac{3}{4}; 0, 0, 0, -2; 0, 0, 0, 0, 1, 0, -3, 0\right) \tag{35}$$

has objective value $5/16$. Collecting its coefficient on each of the sixteen primal variables gives

$$A^T y = \begin{pmatrix} 1 & 1 & 1 & -1 \\ 0 & 0 & 0 & -2 \\ 0 & -1 & -1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix},$$

whereas the cost matrix is

$$C = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix}.$$

Thus $A^T y \leq C$ entrywise, so the dual is feasible. Weak duality yields a lower bound $5/16$, matching (34). □

The scalar two-symbol laws are

$$\mu_P^{*2} = \frac{1}{16}\delta_4 + \frac{6}{16}\delta_6 + \frac{9}{16}\delta_8, \tag{36}$$

$$\mu_Q^{*2} = \frac{9}{16}\delta_6 + \frac{6}{16}\delta_8 + \frac{1}{16}\delta_{10}. \tag{37}$$

Static matching pairs $6/16$ at level 6 and $6/16$ at level 8, for common mass $12/16$. Hence the static residual profiles are

$$R_P^{\text{stat}} = \frac{1}{16}\delta_4 + \frac{3}{16}\delta_8,$$

$$R_Q^{\text{stat}} = \frac{3}{16}\delta_6 + \frac{1}{16}\delta_{10}.$$

The causal optimizer (34) matches $6/16$ at level 6 but only $5/16$ at level 8. Therefore

$$R_P^{\text{causal}} = \frac{1}{16}\delta_4 + \frac{4}{16}\delta_8,$$

$$R_Q^{\text{causal}} = \frac{3}{16}\delta_6 + \frac{1}{16}\delta_8 + \frac{1}{16}\delta_{10}.$$

Consequently,

$$\boxed{R_P^{\text{causal}} = R_P^{\text{stat}} + \frac{1}{16}\delta_8, \qquad R_Q^{\text{causal}} = R_Q^{\text{stat}} + \frac{1}{16}\delta_8.} \tag{38}$$

In particular,

$$\boxed{R_P^{\text{causal}} - R_Q^{\text{causal}} = R_P^{\text{stat}} - R_Q^{\text{stat}}.} \tag{39}$$

**Corollary 7.2** (A causal obstruction invisible to signed scalar spectra)**.** *The additional causal mismatch* $1/16$ *in Theorem 7.1 leaves the signed scalar residual profile unchanged. Hence the signed information-spectrum difference alone cannot detect all causal matching obstructions.*

This is the first exact local manifestation of the phase sector: the lost resource is ordering information, not another scalar-information atom.

## 8 Phase creation under block doubling

The one-dimensional phase $e_{01} - e_{10}$ is special to two symbols. At larger scales the hidden sector grows rapidly, and block doubling creates new phase directions even if the input blocks are individually scalarized.

Identify

$$V_{2m} \cong V_m \otimes V_m$$

by concatenating two length-$m$ words. Define

$$C_m : \mathbb{R}^{m+1} \otimes \mathbb{R}^{m+1} \to \mathbb{R}^{2m+1}$$

on basis vectors by

$$C_m(e_i \otimes e_j) = e_{i+j}. \tag{40}$$

Then the scalarization maps satisfy the exact factorization

$$\boxed{\Pi_{2m} = C_m \circ (\Pi_m \otimes \Pi_m).} \tag{41}$$

Let

$$K_m := \ker \left( C_m|_{\Pi_m(U_m) \otimes \Pi_m(U_m)} \right), \tag{42}$$

and identify this kernel back inside $U_m \otimes U_m$ using the isomorphism $\Pi_m|_{U_m}$.

**Theorem 8.1** (Exact phase decomposition under doubling)**.** *For every $m \geq 1$,*

$$\boxed{W_{2m} = (W_m \otimes W_m) \oplus (W_m \otimes U_m) \oplus (U_m \otimes W_m) \oplus K_m.} \tag{43}$$

*Moreover,*

$$\boxed{\dim K_m = m^2.} \tag{44}$$

*Proof.* From (26),

$$V_m \otimes V_m = (U_m \otimes U_m) \oplus (U_m \otimes W_m) \oplus (W_m \otimes U_m) \oplus (W_m \otimes W_m).$$

Equation (41) annihilates every tensor component containing a factor from $W_m$. On $U_m \otimes U_m$, it is identified with $C_m$. Therefore the kernel is exactly the direct sum in (43).

The map $C_m$ is surjective because every basis vector $e_k$, $0 \leq k \leq 2m$, equals $C_m(e_i \otimes e_j)$ for some $0 \leq i, j \leq m$ with $i + j = k$. Hence

$$\dim K_m = (m+1)^2 - (2m+1) = m^2.$$

□

The sector $K_m$ has a simple interpretation. Even if each half-block has already been reduced to its Hamming layer, the total layer $i + j$ forgets how the weight was split between the two halves. Thus $K_m$ is a genuinely *fresh anti-diagonal phase sector*: it is created by coarse-graining itself, not inherited from unresolved phase inside either sub-block.

### 8.1 The first doubling: two symbols to four

For $m = 2$, $\dim W_2 = 1$ and $\dim U_2 = 3$. Hence

$$\dim W_4 = 1 + 3 + 3 + 4 = 11,$$

in agreement with $2^4 - 5 = 11$. Seven dimensions are inherited:

$$\theta \otimes \theta, \qquad \theta \otimes u_i, \qquad u_i \otimes \theta, \qquad i = 0, 1, 2.$$

A convenient basis of the four-dimensional fresh sector $K_2$ is

$$\begin{aligned}
\kappa_1 &= u_0 \otimes u_1 - u_1 \otimes u_0, \\
\kappa_2 &= u_0 \otimes u_2 - u_2 \otimes u_0, \\
\kappa_3 &= u_1 \otimes u_2 - u_2 \otimes u_1, \\
\kappa_4 &= u_0 \otimes u_2 + u_2 \otimes u_0 - 2u_1 \otimes u_1.
\end{aligned}$$

Each vector has zero anti-diagonal sum under $C_2$.

**Corollary 8.2** (No globally one-dimensional phase model)**.** *The phase direction* $01 - 10$ *cannot close under block doubling. Any exact scale description retaining all ordering information must allow additional phase directions, including a fresh* $m^2$*-dimensional sector at scale* $m \to 2m$*.*

This corollary is kinematic: it concerns the ambient phase space. A decisive open problem is to determine which part of this rapidly growing kernel is actually populated by the *positive causal feasibility constraints*. That smaller object will be called the causal phase fiber.

## 9 The cubic defect as a spectral normal form

Let $E$ be the forward shift on test functions,

$$(Ef)(t) := f(t+1).$$

For the dyadic pair define

$$P^* := \frac{1}{4}E^2 + \frac{3}{4}E^4, \qquad Q^* := \frac{3}{4}E^3 + \frac{1}{4}E^5.$$

**Theorem 9.1** (Exact cubic shift factorization)**.**

$$\boxed{P^* - Q^* = \frac{1}{4}E^2(I - E)^3.} \tag{45}$$

*Equivalently,*

$$\boxed{\mu_P - \mu_Q = \frac{1}{4}\delta_2 * (\delta_0 - \delta_1)^{*3}.} \tag{46}$$

*Proof.* Expand $\frac{1}{4}E^2(I - E)^3$. The coefficients are $\frac{1}{4}(1, -3, 3, -1)$ on shifts $2, 3, 4, 5$, exactly those of $P^* - Q^*$. □

Thus every polynomial of degree at most two is annihilated. On the cubic,

$$(P^* - Q^*)t^3 = -\frac{3}{2}. \tag{47}$$

The first transverse polynomial jet is therefore one-dimensional modulo quadratics.

More generally, if a finite signed measure $\nu = \sum_k \nu_k \delta_k$ has vanishing raw moments through order $r-1$, its generating polynomial $V(z) = \sum_k \nu_k z^k$ is divisible by $(1-z)^r$. This is simply the invertible triangular relation between raw moments and falling-factorial moments. On test functions, the associated difference operator factors by $(I-E)^r$.

Laplace characters

$$f_a(t) := e^{-at}$$

are eigenfunctions of the shift, $Ef_a = e^{-a} f_a$. Therefore

$$\boxed{(P^* - Q^*)f_a = \frac{1}{4}e^{-2a}(1-e^{-a})^3 f_a.} \tag{48}$$

The transform used in Section 3 is thus a spectral character of the exact translation algebra, not an unrelated trick.

**Lemma 9.2** (Shift invariance of viability). *If $B \in \mathcal{K}_n$, then $z^s B \in \mathcal{K}_n$ for every integer $s \geq 0$.*

*Proof.* Multiply (16) at every node of a feasible tree by $z^s$. □

*Remark* 9.3 (Why Laplace modes are multipliers, not bill coordinates). If $a > 0$ and $\mathcal{K}_n \neq \varnothing$, then

$$\ell_n(e^{-a\cdot}) = 0.$$

Indeed, shift any fixed feasible law by $s$: its pairing with $e^{-a\cdot}$ is multiplied by $e^{-as}$ and therefore tends to zero as $s \to \infty$; nonnegativity gives the reverse inequality. Laplace modes are therefore useful as dual multipliers and shift eigencharacters, not as direct coordinates of $\mathcal{B}_n = \ell_n(\mathrm{id})$.

### 9.1 Why the third order is critical for this converse

Suppose more generally that

$$d(a) := -\log \rho(a) = c_r a^r + O(a^{r+1}), \qquad c_r > 0,$$

while $\log \Lambda(a) = \sigma^2 a^2 + O(a^3)$. Then

$$\Gamma(a) = 2a + \frac{\sigma^2}{c_r} a^{3-r}(1 + o(1)). \tag{49}$$

For this proof mechanism, $r = 3$ is critical: $r = 2$ makes the logarithmic scale too small, $r = 3$ gives a finite nonzero coefficient, and $r > 3$ requires a different scaling or a sharper dual. This is a statement about the present converse mechanism, not a universal exponent theorem.

## 10 Green coordinates and boundary renormalization

Set

$$P(z) := 1 + 3z^2, \qquad A(z) := z(3 + z^2).$$

Then

$$\boxed{P(z) - A(z) = (1-z)^3.} \tag{50}$$

The original PGFs are regular at $z = 1$; the analytic difficulty is the triple zero. We resolve it by dividing by the corresponding Green kernel.

**Definition 10.1** (Cubic Green potential). For a buffer PGF $B$, define

$$\boxed{T_B(z) := \frac{1 - B(z)}{(1-z)^3}.} \tag{51}$$

**Theorem 10.2** (Exact affine Green conjugacy)**.** *The one-step zero-repair recursion* (16) *is equivalent to*

$$\boxed{T_{B_0}(z) + 3z^2 T_{B_1}(z) = A(z)T_B(z) + 1.} \tag{52}$$

*Proof.* Using $P - A = (1-z)^3$,

$$\begin{aligned}(1 - B_0) + 3z^2(1 - B_1) &= P - AB \\ &= A(1 - B) + (P - A).\end{aligned}$$

Divide by $(1-z)^3$. □

Thus the signed cubic source becomes the positive unit source. Positivity of the probability cone, rather than inversion of the cubic difference, is the remaining barrier.

Let $\beta_n := \mathbb{P}_B(J > n)$ and $T_B(z) = \sum_{n\geq 0} t_n z^n$. Direct coefficient extraction gives

$$t_n = \sum_{k=0}^{n} (n - k + 1)\beta_k. \tag{53}$$

With the backward difference $\nabla t_n = t_n - t_{n-1}$,

$$\boxed{\nabla t_n = \sum_{k=0}^{n} \beta_k, \qquad \nabla^2 t_n = \beta_n, \qquad \nabla^3 t_n = -\mathbb{P}(J = n)\ (n \geq 1).} \tag{54}$$

In particular,

$$\boxed{\mathbb{E}_B J = \sum_{n\geq 0} \beta_n = \lim_{n\to\infty} \nabla t_n.} \tag{55}$$

The bill is therefore an asymptotic slope in Green coordinates.

Iterating (52) for $m$ generations gives

$$\sum_{w\in\{0,1\}^m} 3^{\mathrm{wt}(w)} z^{2\,\mathrm{wt}(w)} T_{B_{hw}}(z) = A(z)^m T_{B_h}(z) + G_m(z), \tag{56}$$

where

$$\boxed{G_m(z) = \frac{P(z)^m - A(z)^m}{(1-z)^3} = \sum_{j=0}^{m-1} P(z)^{m-1-j} A(z)^j.} \tag{57}$$

Hence $G_m$ is coefficientwise nonnegative. At $m = 2$,

$$\boxed{G_2(z) = P(z) + A(z) = (1 + z)^3.} \tag{58}$$

The normalized mask $(1 + z)^3/4$ is the standard binary quadratic B-spline subdivision mask (B-spline order three). The connection is exact at the algebraic level: subdivision theory relates such masks, vanishing moments and polynomial reproduction [6].

Define the positive boundary dilation

$$\boxed{(\mathcal{R}F)(z) := \frac{1}{4}(1 + z)^3 F(z^2).} \tag{59}$$

This is not asserted to be the full nonlinear Bellman renormalization. It is the exact scale map in the cubic Green boundary algebra.

**Proposition 10.3** (Exact third-difference conjugacy)**.** *Let* $\Delta_3 F := (1-z)^3 F$ *and* $(D_2G)(z) := G(z^2)$*. Then*

$$\boxed{\Delta_3\mathcal{R} = \frac{1}{4}D_2\Delta_3.} \tag{60}$$

*Proof.* Use $(1-z)^3(1+z)^3 = (1-z^2)^3$. □

**Proposition 10.4** (Boundary pole spectrum)**.** *If*

$$F(z) = \frac{c+o(1)}{(1-z)^r} \qquad (z \uparrow 1),$$

*then*

$$\boxed{(\mathcal{R}F)(z) = 2^{1-r}\frac{c+o(1)}{(1-z)^r}.} \tag{61}$$

Thus pole orders $r = 0, 1, 2$ have multipliers $2, 1, 1/2$, respectively. The word "spectrum" here refers only to the exact boundary scaling law (61); no spectrum of the full viability operator has yet been proved.

# 11 An explicitly controllable marginal phase channel

A two-block causal correction arising in the dyadic model is the signed packet

$$\boxed{\eta(z) = \frac{1}{16}(z^2+z^4-1-z^6) = -\frac{1}{16}(1-z^2)(1-z^4).} \tag{62}$$

It is mass- and mean-neutral:

$$\eta(1) = 0, \qquad \eta'(1) = 0,$$

while

$$\eta''(1) = -1. \tag{63}$$

Under the signed residual dilation

$$(\mathcal{S}\Theta)(z) := \frac{1}{4}\Theta(z^2),$$

the factorial moments $M_j = \Theta^{(j)}(1)$ transform as

$$\begin{aligned} M_0 &\mapsto \frac{1}{4}M_0, \\ M_1 &\mapsto \frac{1}{2}M_1, \\ M_2 &\mapsto M_2 + \frac{1}{2}M_1, \\ M_3 &\mapsto 2M_3 + 3M_2. \end{aligned}$$

Hence $M_2$ is exactly marginal on the subspace $M_0 = M_1 = 0$.

Apply the cubic Green transform to $\eta$:

$$\boxed{\Phi(z) := -\frac{\eta(z)}{(1-z)^3} = \frac{(1+z)^2(1+z^2)}{16(1-z)}.} \tag{64}$$

Its coefficients are

$$[z^n]\Phi = \frac{1}{16}(1, 3, 5, 7, 8, 8, 8, \ldots),$$

so $\Phi$ lies in the simple-pole marginal sector of (61) and carries zero asymptotic mean slope.

**Proposition 11.1** (Positive cohomological defect)**.**

$$\boxed{\Phi - \mathcal{R}\Phi = \frac{(1+z)^3(1+z^2)(z^4+2z^2+3)}{64}.} \tag{65}$$

*The right-hand side has nonnegative coefficients. Consequently,*

$$0 \le \mathcal{R}^k\Phi \le \Phi \qquad (k \ge 0) \tag{66}$$

*coefficientwise.*

*Proof.* Direct substitution into (59) gives (65). Positivity of $\mathcal{R}$ then allows iteration. □

Define

$$\eta_s(z) := 4^{-s}\eta(z^{2^s}), \qquad s \ge 0,$$

and

$$\boxed{E_*(z) := \frac{2}{3}z^2 + \sum_{s\ge 0} \frac{4^{-s}}{16}\left(1 + z^{2^{s+1}} + z^{2^{s+2}} + z^{6\cdot 2^s}\right).} \tag{67}$$

**Theorem 11.2** (Universal finite-mean envelope for the explicit phase channel)**.** *The series $E_*$ is a probability PGF with*

$$\boxed{\mathbb{E}_{E_*} J = \frac{17}{6}.} \tag{68}$$

*For every sequence $(\theta_s)_{s\ge 0}$ with $|\theta_s| \le 1$,*

$$E_*(z) + \sum_{s\ge 0} \theta_s \eta_s(z) \tag{69}$$

*is again a probability PGF with the same mean* $17/6$*.*

*Proof.* The envelope part has total mass

$$\sum_{s\ge 0} \frac{4\cdot 4^{-s}}{16} = \frac{1}{3},$$

which together with $2/3$ from the first atom gives mass one. The first term contributes mean $4/3$, while scale $s$ contributes

$$\frac{4^{-s}}{16}\left(2^{s+1} + 2^{s+2} + 6\cdot 2^s\right) = \frac{3}{4}\,2^{-s}.$$

Summing yields $4/3 + 3/2 = 17/6$. Coefficientwise, $E_*$ dominates the sum of absolute masses of every packet that can contribute at that coefficient; the triangle inequality gives nonnegativity of (69). Each $\eta_s$ has zero mass and zero first moment, so normalization and mean are unchanged. □

*Remark* 11.3 (The limit of this theorem)*.* Theorem 11.2 controls one explicit phase channel through arbitrarily many scales at $O(1)$ mean cost. It does not show that the full causal phase fiber decomposes into copies of this channel. Theorem 8.1 in fact warns that fresh phase dimensions are created at every blocking step.

## 12 A critical self-similar reservoir

The dilation algebra also contains a canonical Mahler fixed point. Define

$$(\mathfrak{M}B)(z) := \frac{3}{4}z^2 + \frac{1}{4}B(z^2). \tag{70}$$

**Proposition 12.1** (Mahler fixed point)**.** *There is a unique probability PGF satisfying $B = \mathfrak{M}B$, namely*

$$\boxed{B_*(z) = \sum_{k\geq 1} \frac{3}{4^k} z^{2^k}.} \tag{71}$$

*It has*

$$\mathbb{E}_{B_*} J = 3, \qquad \mathbb{E}_{B_*} J^2 = \infty. \tag{72}$$

*Proof.* Iterating $B(z) = \frac{3}{4}z^2 + \frac{1}{4}B(z^2)$ determines the coefficients and gives (71); its masses sum to one. The first moment is

$$\sum_{k\geq 1} \frac{3}{4^k} 2^k = 3,$$

while the second moment is

$$\sum_{k\geq 1} \frac{3}{4^k} 4^k = \infty.$$

□

The fixed point is therefore exactly critical between finite first and second moments. This fact should not be overinterpreted: a fixed point of the static scale map is not automatically an infinite zero-repair tree, and Theorem 3.5 forbids any such tree with finite root mean. The Mahler law is best viewed as a canonical boundary object of the scale algebra, not as a completed causal construction.

## 13 Positive fibers over scalar information

The quotient $V_m \to \mathbb{R}^{m+1}$ identifies exactly which linear information is lost when one passes from ordered histories to total information. Positivity can also be described exactly before causality is imposed.

For $0 \leq k \leq m$ recall the normalized Hamming-layer vector

$$u_k^{(m)} := \frac{1}{\binom{m}{k}} \sum_{|w|=k} e_w,$$

and define the canonical section

$$\sigma_m(r) := \sum_{k=0}^{m} r_k u_k^{(m)}, \qquad r = (r_0, \dots, r_m) \in \Delta_m. \tag{73}$$

Here $\Delta_m$ denotes the probability simplex on the $m+1$ Hamming layers.

**Theorem 13.1** (Exact positive phase fiber)**.** *For $r \in \Delta_m$, the set of probability laws on ordered histories with scalar profile $r$ is*

$$\boxed{\Pi_m^{-1}(r) \cap \Delta(\{0,1\}^m) = \sigma_m(r) + \mathsf{F}_m(r),} \tag{74}$$

*where*

$$\boxed{\mathsf{F}_m(r)=\left\{\theta\in W_m:\theta_w\geq-\frac{r_{|w|}}{\binom{m}{|w|}}\ \textit{for every } w\right\}.}\tag{75}$$

*Equivalently, the fiber is the direct product, over Hamming layers, of scaled simplices of total mass $r_k$. If every $r_k>0$, its affine dimension is*

$$\sum_{k=0}^{m}\left(\binom{m}{k}-1\right)=2^m-(m+1).$$

*Proof.* If $\nu$ is a probability vector with $\Pi_m\nu=r$, then the total mass of $\nu$ on the layer $|w|=k$ is $r_k$. Since $\sigma_m(r)$ places the uniform mass $r_k/\binom{m}{k}$ on that layer, the difference $\theta=\nu-\sigma_m(r)$ has zero sum on each layer, hence $\theta\in\ker\Pi_m=W_m$. The condition $\nu_w\geq0$ is exactly (75). The converse is immediate. The product-of-simplices statement follows layer by layer, and the dimension formula follows by summing the dimensions $\binom{m}{k}-1$. □

Theorem 13.1 is the exact analogue of a coefficientwise positivity test at the *ambient* history level. It is not yet the causal criterion we ultimately need. Prefix compatibility cuts a generally much smaller subset out of this product of simplices. The two-symbol theorem in Section 7 proves that this cut is already strict at the first nontrivial block length.

# 14 Why finite closures cannot be the answer

The phase-space growth above might tempt one to search for a fortunate finite collection of exact buffer profiles or for one scalar potential that absorbs all of the ordering information. Both possibilities are ruled out.

**Theorem 14.1** (No finite exact profile automaton)**.** *There do not exist finitely many probability PGFs $B^{(1)},\dots,B^{(K)}$ and transition maps $\tau_0,\tau_1:\{1,\dots,K\}\to\{1,\dots,K\}$ such that*

$$z(3+z^2)B^{(s)}(z)=B^{(\tau_0(s))}(z)+3z^2B^{(\tau_1(s))}(z)\tag{76}$$

*for every $s$.*

*Proof.* Let $G=(B^{(1)},\dots,B^{(K)})^\top$, and let $L,H$ be the selector matrices induced by $\tau_0,\tau_1$. Then

$$M(z)G(z)=0,\qquad M(z):=z(3+z^2)I-L-3z^2H.$$

The coefficient of $z^{3K}$ in $\det M(z)$ is 1: it is obtained uniquely by selecting the $z^3$ term from every diagonal entry. Thus $\det M$ is not the zero polynomial. For all but finitely many $z\in(0,1)$, $M(z)$ is invertible and therefore $G(z)=0$, contradicting positivity of probability PGFs on $(0,1)$. □

**Theorem 14.2** (No positive scalar state-only drift)**.** *Suppose $\phi:\mathbb{Z}_{\geq0}\to\mathbb{R}$ and constants $c,\lambda_x,\mu_y$ satisfy*

$$\phi(j+y-x)-\phi(j)\geq c+\lambda_x+\mu_y\tag{77}$$

*whenever $j+y-x\geq0$, with gauges*

$$\sum_x P(x)\lambda_x=0,\qquad\sum_y Q(y)\mu_y=0.$$

*For the dyadic pair, $c\leq0$.*

*Proof.* The admissible increment differences contain both $+1$ and $-1$. Applying (77) to these moves bounds the tail first differences $d_j := \phi(j+1) - \phi(j)$ from above and below. Choose a sequence of long intervals along which their Cesàro means converge to a finite number $a$. Average (77) over such an interval, and then over $P \otimes Q$. Boundary terms vanish after division by the interval length, while the gauge terms average to zero. The remaining translation term is $a\mathbb{E}(Y - X)$. Since the increment means agree, it is zero, and hence $c \leq 0$. $\square$

These theorems have a useful positive interpretation. A sharp asymptotic object, if it exists, must retain information at growing scales. The large kernel $W_m$ is not an artifact of a poor coordinate system that can be compressed into finitely many exact modes without loss.

# 15 The renormalized positive phase problem

We can now state the unresolved problem without appealing to a particular buffer ansatz.

The linear quotient $V_m = U_m \oplus W_m$ separates visible total-information profiles from ordering information. Theorem 13.1 describes the complete positive fiber before causality. Causality adds prefix constraints. Let

$$\mathsf{C}_m(r) \subseteq \mathsf{F}_m(r) \tag{78}$$

denote, schematically, the subset of phase perturbations that occur in a prefix-compatible positive lift with the prescribed coarse data $r$ and the relevant future viability constraints. The notation is intentionally abstract: identifying a useful intrinsic description of $\mathsf{C}_m(r)$ is part of the problem, not an assumption hidden in the notation.

The role of Section 8 is now clear. Blocking from $m$ to $2m$ does not merely transport an old phase vector. It creates the fresh sector $K_m$ of dimension $m^2$. Thus any exact scale map must account for at least three operations:

$$\boxed{\begin{array}{c}\text{transport inherited phase} + \text{create inter-block phase}\\ + \text{ project back into the positive causal fiber.}\end{array}}$$

The last operation is nonlinear because positivity and future viability are inequalities.

## 15.1 The central characterization problem

The closest analogue, in the present problem, of a complete coefficientwise-domination theorem is the following.

**Open Problem 15.1** (Causal phase-fiber characterization)**.** *Give necessary and sufficient conditions, stated intrinsically in coarse and phase coordinates, for a positive block profile to admit a prefix-compatible zero-repair lift through one scale. Equivalently, characterize the sets $\mathsf{C}_m(r)$ by explicit inequalities or by an exact positive factorization that is stable under blocking.*

For $m = 2$, Proposition 6.3 is an exact coordinate form of this question: the unknown signed series $\Theta$ must lie simultaneously in four positivity regions. For general $m$, Theorem 8.1 shows why a single $\Theta$ cannot suffice.

## 15.2 A conditional scale theorem

The following simple implication isolates exactly what an upper-bound theorem would have to deliver.

**Proposition 15.2** (Bounded scale repair implies logarithmic growth). *Suppose there is a constant $C < \infty$ such that*

$$\boxed{\mathcal{B}_{2n} \le \mathcal{B}_n + C} \tag{79}$$

*for every $n \ge 1$. Then*

$$\mathcal{B}_n = O(\log n). \tag{80}$$

*For the dyadic parity pair, this combines with Theorem 4.1 to give*

$$\boxed{\mathcal{B}_n = \Theta(\log n).} \tag{81}$$

*Proof.* Iterating (79) gives $\mathcal{B}_{2^k} \le \mathcal{B}_1 + kC$. Monotonicity of the finite-horizon viability bill then gives the same order for arbitrary $n$ by choosing $2^{k-1} < n \le 2^k$. The lower bound is Theorem 4.1. $\square$

Equation (79) is not proved here. It is the most concrete sharp upper target suggested by the scale algebra. If the dyadic increment converged,

$$\mathcal{B}_{2n} - \mathcal{B}_n \longrightarrow \gamma,$$

then along dyadic horizons one would obtain

$$\mathcal{B}_n \sim \frac{\gamma}{\log 2} \log n,$$

and the rigorous lower bound would force

$$\gamma \ge \frac{\log 2}{3}. \tag{82}$$

No equality is asserted.

### 15.3 Why the one-channel Jordan picture is insufficient

The explicit packet $\eta_s$ is marginal in the second factorial moment, but its first-moment carrying cost decays geometrically: an envelope mass of order $4^{-s}$ transported to distance $2^s$ costs order $2^{-s}$. This is exactly why Theorem 11.2 has finite mean. Consequently, one copy of the explicit $\eta$ channel cannot by itself explain logarithmic growth.

A more plausible critical mechanism, if the logarithmic lower bound is sharp, is the repeated creation of fresh phase directions under blocking. The ambient fresh sector has dimension $m^2$, while the cubic signed residual is suppressed by a third-order scale factor. The dimensional balance

$$m^2 \cdot m^{-3} \cdot m \asymp 1 \tag{83}$$

is suggestive: multiplicity, cubic residual mass, and first-moment transport distance cancel at order three. We emphasize that (83) is a scaling heuristic, not a theorem. It does not show that $\Theta(m^2)$ feasible phase directions are independently excited, nor that positivity prevents cancellations. Its value is that it identifies a falsifiable mechanism compatible with both the exact cubic symbol and the exact growth of the phase kernel.

## 16 Relation to neighboring mathematical theories

**Adapted and causal optimal transport.** The causal constraint in Section 7 is a standard nonanticipativity constraint on a coupling. General causal and bicausal transport duality, including dual attainment under broad measurable-cost hypotheses, is available in [9]. Very recent graph-causal optimal transport characterizes gluing properties and dynamic programming for causality prescribed by directed acyclic graphs [16]. These results make adapted transport a natural surrounding language. They do not make the present problem automatic: exact inventory conservation and the hard cone constraint $J \geq 0$ must still be incorporated, and the full countable-state no-gap theorem for our Bellman–Fenchel recursion has not been proved.

**Convex dynamic programming.** The passage from convex viability sets to lower support functionals is a finite-dimensional convex duality calculation, not a new general duality theorem. For infinite-dimensional stochastic optimization with information constraints, the recent work of Pennanen and Perkkiö provides a broader convex-duality and dynamic-programming framework [11, 12]. The unresolved issue here is to choose a weighted function/measure topology in which the unbounded mean objective, shift invariance and positivity coexist with strong duality.

**Nonlinear Perron–Frobenius theory.** Order-preserving homogeneous and additively homogeneous maps, quotienting by constants, and projective metrics are standard in nonlinear Perron–Frobenius theory and in Shapley operators [5, 8, 7]. Recent work also extends joint spectral-radius ideas to switched nonlinear order-preserving systems [10]. The relevance here is structural, not a theorem transfer: Theorem 14.1 already rules out an exact finite-mode state, and the admissible phase fiber is infinite-dimensional and constrained by positivity.

**Subdivision and vanishing moments.** The identity

$$\Delta_3 \mathcal{R} = \frac{1}{4} D_2 \Delta_3$$

places the boundary algebra directly next to subdivision schemes. The mask $(1+z)^3/4$ is the binary quadratic B-spline mask; its order-three zero condition is the same algebra that annihilates the matched mass, mean and variance jets. Sum rules, polynomial reproduction and invariant difference spaces are classical tools in subdivision theory [6]. The novelty of the present difficulty is the additional causal-positive fiber sitting over this linear scheme.

**Finitary coding and randomness recycling.** The zero-repair equation resembles a constrained information coboundary, while finitary isomorphism theory shows that stronger moment assumptions on coding length preserve increasingly fine information invariants [2, 3, 4]. In a different online-sampling model, modern randomness-recycling algorithms can approach the Shannon limit using persistent state; recent real-probability algorithms achieve expected entropy $\sum_i H(F_i) + O(\log n)$ with $O(\log n)$ persistent space [13, 14]. Neither result implies a zero-repair theorem, but both reinforce the need to distinguish entropy rate from persistent working state.

**Synchronization-channel duality.** Kovačević's recent exact sticky-channel calculation is a useful nearby example of proof architecture rather than a source of a theorem used here: a channel is reduced to a simpler operational object, a dual bound is made exact by a coefficientwise criterion, and the entire domination class is then characterized algebraically [15]. Problem 15.1 asks for the analogous structural step for a causal-positive phase fiber.

## 17 Back to "Who pays the bill?"

The original resource question was broader than the zero-repair model. An observable stochastic system may be realizable at the ordinary Shannon rate while different exact-realization contracts impose different requirements on its hidden state. The mathematics above separates three quantities that should not be conflated:

$$\boxed{\text{entropy rate} \quad \neq \quad \text{fresh repair expenditure} \quad \neq \quad \text{exact causal working inventory.}} \tag{84}$$

The first is a rate: how many new random bits per step are required asymptotically. The second measures information that a simulator is permitted to inject after seeing where mismatch occurred. The third is the present quantity: a nonnegative inventory that must make every one-step information identity exact without such fresh repair.

For the dyadic parity pair, the present paper proves a sharp qualitative statement about the third resource. Even though mass, mean and variance of the two information laws agree, their third-order mismatch cannot be carried forever by any finite-mean zero-repair inventory:

$$\mathcal{B}_n \geq \left(\frac{1}{3} - o(1)\right) \log n. \tag{85}$$

At the same time, the two-symbol causal matching theorem shows why a scalar information spectrum alone is insufficient: causality creates an additional positive obstruction while leaving the signed scalar residual unchanged. Thus part of the bill is carried by ordering information that disappears under the usual scalar projection.

This does *not* prove that an unrestricted causal simulator pays an additional asymptotic entropy rate. In the broader causal-tree model, fresh randomness, nonlocal blocks and hidden dependence may implement strategies forbidden by zero repair. A companion analysis of that broader contract finds that, for two finite actions, the first-order causal rate can reach the observable Shannon floor [17]. There is no contradiction: rate and working inventory are different resources.

The missing bridge is therefore mathematically precise.

**Open Problem 17.1** (Entropy–inventory comparison)**.** *Establish the sharpest general comparison, if any, between the finite-horizon unrestricted causal entropy residual and the zero-repair viability bill $\mathcal{B}_n$. In particular, determine which forms of fresh repair or blockwise hidden dependence can convert a diverging working-inventory requirement into only $o(n)$ additional source entropy.*

The answer would return the phase-renormalization theory developed here to the original accounting problem. Until such a bridge is proved, the correct conclusion is operational rather than ontological: exact realization costs depend on what must remain simultaneously valid, on what may be repaired later, and on which history information the hidden state is allowed to retain.

## 18 Theorem ledger and open frontier

For clarity, Table 1 separates the proved statements from the renormalization program.

The frontier can be summarized in one sentence:

| |
|---|
| the cubic linear defect is explicitly solvable;<br>the positive causal phase fiber is the remaining object. |

## 19 Conclusion

The first nontrivial dyadic zero-repair problem contains two distinct kinds of information. The scalar information profile records how much surprisal has accumulated; the phase fiber records in which order that information was created. Static convolution sees only the first. Causality sees both.

This distinction is already measurable after two symbols. The static mismatch is $1/4$, but the causal mismatch is exactly $5/16$. The additional $1/16$ appears identically on both scalar residuals and is therefore invisible to their signed difference. At larger block lengths the hidden space expands rapidly: $W_m$ has dimension $2^m - (m+1)$, and every doubling creates a fresh $m^2$-dimensional anti-diagonal sector even before positivity is imposed.

| Statement | Status |
|---|---|
| Finite-horizon fixed-tilt logarithmic converse | proved |
| Third-cumulant lower bound $\liminf \mathcal{B}_n/\log n \geq \Delta\kappa_3/(6\sigma^2)$ | proved |
| Dyadic lower coefficient $1/3$ and $\mathcal{B}_n \to \infty$ | proved |
| Finite-dimensional Bellman–Fenchel duality | proved |
| Countable-state Bellman formula as weak dual certificate | proved |
| Phase quotient $W_m = \ker \Pi_m$, $\dim W_m = 2^m - (m+1)$ | proved |
| Exact positive scalar-profile fiber, Theorem 13.1 | proved |
| Two-symbol causal mismatch $5/16$ with exact primal and dual certificates | proved |
| Causal surcharge invisible to the signed scalar residual | proved |
| Block-doubling decomposition of $W_{2m}$ and fresh sector $\dim K_m = m^2$ | proved |
| Cubic shift factorization and Green affine conjugacy | proved |
| Explicit marginal phase cohomology and mean-17/6 envelope | proved* |
| Mahler scale fixed point with finite first and infinite second moment | proved |
| Finite exact profile automaton | impossible |
| Positive scalar state-only Bellman drift | impossible |
| Intrinsic characterization of the causal-positive phase fiber $\mathsf{C}_m(r)$ | open |
| Strong countable-state Bellman–Fenchel duality in a natural weighted space | open |
| Uniform doubling bound $\sup_n(\mathcal{B}_{2n} - \mathcal{B}_n) < \infty$ | open |
| Matching $O(\log n)$ upper bound / $\Theta(\log n)$ law | open |
| Exact asymptotic constant for $\mathcal{B}_n/\log n$ | open |
| General comparison with unrestricted causal entropy residual | open |

Table 1: Logical status of the main statements. The symbol * means: proved for the explicit phase channel stated in Section 11, not for the full causal phase fiber.

Independently, the source mismatch has an exact cubic normal form. The factor $(I - E)^3$ annihilates the matched mass, mean and variance jets; division by the corresponding cubic zero produces a Green coordinate with a positive unit source and a quadratic B-spline boundary mask. The same order three is critical in the rigorous Laplace converse, which forces

$$\mathcal{B}_n \geq \left(\frac{1}{3} - o(1)\right) \log n$$

for the dyadic pair.

The two structures meet at the unresolved point. Linear cubic inversion is not difficult; keeping all descendants inside the positive, prefix-compatible phase fiber is. A complete characterization of that fiber, stable under scale doubling, would turn the present renormalization picture into an actual upper-bound mechanism. A bounded cost per dyadic scale would close the growth order at $\Theta(\log n)$.

The broader information-theoretic lesson is narrower but already rigorous: an exact stochastic realization can have different costs depending on whether one prices source entropy, fresh repair, or nonnegative causal working inventory. The scalar information spectrum does not contain all of the causal bookkeeping. Ordering information can carry a bill of its own.

## Research-status and reproducibility note

All asymptotic claims in this paper are supported by analytic arguments. The exact two-symbol causal value is accompanied by explicit rational primal and dual certificates, so it does not depend on floating-point optimization. Computer algebra was used to check finite polynomial identities and exact coefficients during development, not as a substitute for the stated proofs.

The paper deliberately uses conservative priority language. The surrounding subjects—adapted transport, finitary coding, nonlinear positive operators, subdivision theory and synchronization channels—are mature, and an equivalent formulation of some structural lemma may exist elsewhere. Where a direct precedent is known, it is cited; no stronger historical novelty claim is intended.

During preparation, the author used OpenAI's ChatGPT as a research assistant for literature discovery, algebraic checking, adversarial proof review and LaTeX preparation. The mathematical statements are separated explicitly into proved results and open problems in Table 1.

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